Unit 1, Lesson 3: Interpreting Function Notation in the Real World

Benchmark

MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.

Learning Target

- ☐ I can evaluate a function for a specific input value of the domain and interpret it in a real-world context.

1.3.1 Warm-Up: Equivalent Function Values

1. Fill in the boxes to construct two functions where the final equality statement is true. Use the digits 1 to 9 once each at most.

$$g(x) = \square\, x - \square$$

$$g(\square) = 26$$

1.3.2 Guided Instruction: Interpreting Functions

Recall that functions have a special notation which allows unique relationships to be expressed in a concise and meaningful way.

The value of a function is dependent on the input. The statement $D(2) = 60$ means that for an input of 2, the value of the function, D, is 60.

Functions can be used to model real-world relationships.

1. If $F(t)$ models the outside temperature in degrees Fahrenheit t hours after 6 a.m., the statement $F(2) = 60$ has a specific meaning within the context.

   $F(2) = 60$ can be interpreted as ______ hours after ____ a.m., which is ______ a.m., the outside temperature is ______ degrees Fahrenheit.

2. Physics students drop a ball from the top of a 400 foot high building and model its height above the ground as a function of time using the function $H(t) = 400 - 16t^2$. The height, H, is measured in feet and time, t, is measured in seconds.

   a. Find the value of $H(0)$.

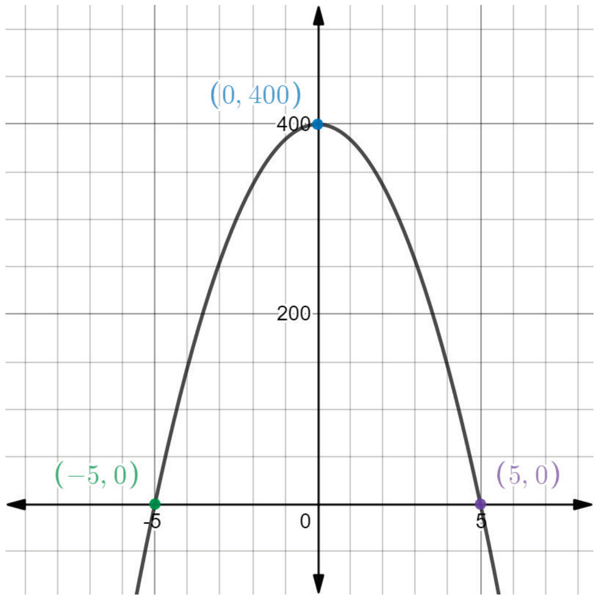
   b. What does $H(0)$ mean in the context of this problem?

   c. Evaluate $H(2)$.

   d. Interpret the meaning of $H(2)$ based on the given context.

   e. José is asked to interpret the value of $H(-3)$. Is interpreting this value reasonable in the context of this situation?

3. The graph of $H(t)$ is shown on the coordinate grid.



a. Based on the context, the x-axis represents _______________ and the y-axis represents ________________.

b. What will the height of the ball be when it hits the ground?

c. Circle the point where the ball will hit the ground.

d. Write the point where the ball hits the ground using function notation.

$$H(\_\_\_\_) = \_\_\_\_\_$$

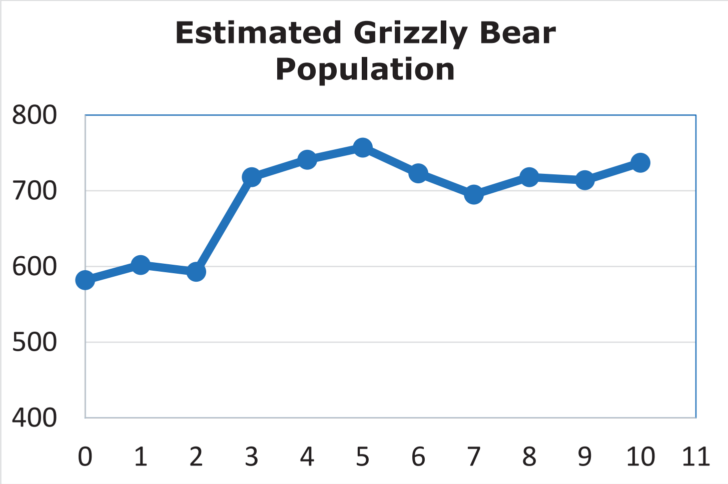
e. How long did it take the ball to hit the ground after it was released?

f. Discuss with your partner what $H(1.8) > H(3.4)$ means in terms of the context.

1.3.3 Guided Practice: Interpreting Functions

1. The population of grizzly bears at Yellowstone National Park has been threatened with local extinction since 1975. Through many conservation efforts, grizzlies have made a remarkable recovery. The function $y = G(t)$ is modeled below, where t is the number of years since 2009 and G is the number of grizzly bears.

   a. Label the x and y axes based on the context.



   b. Find the approximate value from the graph above.

$$G(3) \approx \_\_\_\_\_\_$$

   c. In the context of this problem, $G(3)$ means that 3 years after 2009, there were approximately _______ bears in Yellowstone.

   d. Find the approximate value from the graph.

$$G(8\,½) \approx \_\_\_\_\_\_$$

   e. What does $G(8\,½)$ mean in the context of this problem?

   f. Explain why there is not a value of t, such that $G(t) = 710.5$.

2. Matthew filled his bathtub, took a bath, and then drained the tub. The function $B(m)$ gives the depth of the water, in inches, m minutes after he began to fill the bathtub.

   a. Explain the meaning of each statement in this context.

   $B(0) = 0$

   $B(9.5) = B(20)$

   b. Discuss with your partner why the inequality $B(1) < B(7)$ is likely true in this context.

3. A biologist is studying the relationship between a tree's diameter and its height. She records the following data for seven different trees using the function $H(d)$, where H represents the height, in feet, and d represents the diameter, in inches.

Diameter (inches)	2	4	6	8	10	12	14
Height (feet)	6	7	10	15	18	22	28

   a. Fill in the blanks in the following sentence to interpret the statement $H(10) = 18$ in this context.

   A tree with a diameter of ______ has a height of ______.

   b. Determine the value of $H(14)$.

   c. Interpret $H(14)$ for the given context.

1.3.4 Try It: Evaluating and Interpreting Functions

1. For groups of 80 or more people, a charter bus company determines the rate per person according to the formula $R(n) = 8 - 0.05(n - 80)$, where n represents the number of people and R represents the rate, in dollars, charged to each customer.

   a. Find each value and interpret it in context.

   $R(75) =$ ______

   $R(85) =$ ______

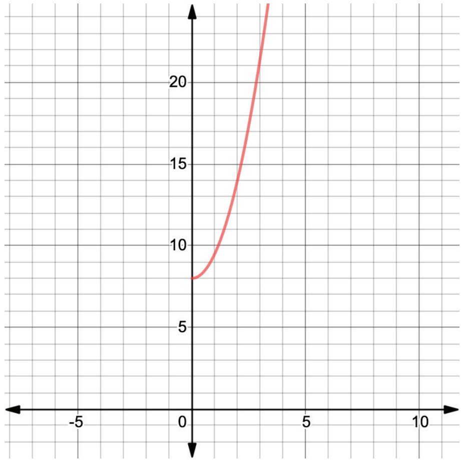
   b. Why is $R(75) > R(85)$ in this context?

1.3.5 Your Turn!

1. The size of a fungal colony is modeled by the equation $z(t) = \frac{3}{2}t^2 + 8$, where z is the size of the colony in microns, and t is the time, in hours.

   A graph of the function $z(t)$ is shown.

   Using the graph, approximate the values and interpret their meaning in this context.



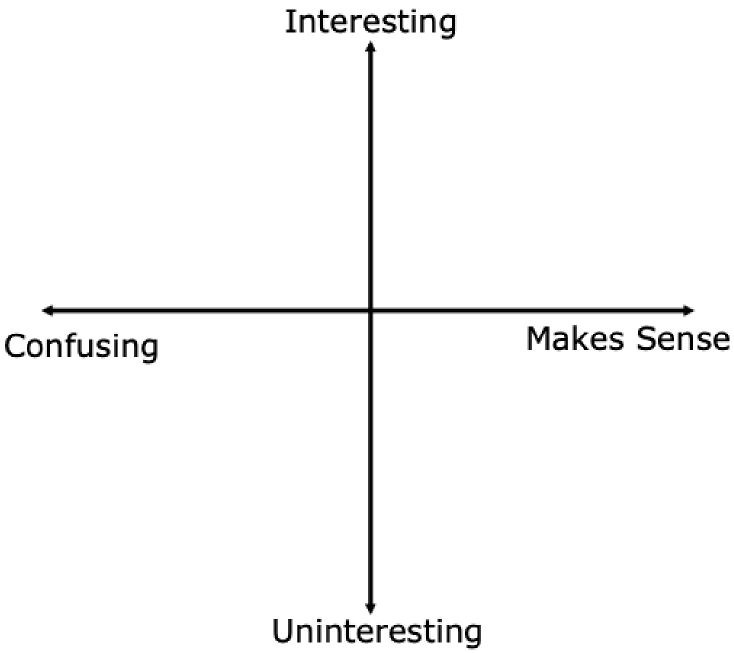
   a. $z(3) =$ ______

   b. $z(0) =$ ______

   c. Why does the graph only display y-values for the function when $t \geq 0$?

1.3.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate where you feel you are on today's learning target: "I can evaluate a function for a specific input value of the domain and interpret it in a real-world context."



Unit 1, Lesson 4: Applying the Laws of Exponents

Benchmark

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

Learning Targets

- ☐ I can apply the Laws of Exponents to write equivalent numerical expressions.
- ☐ I can evaluate square roots and cube roots.